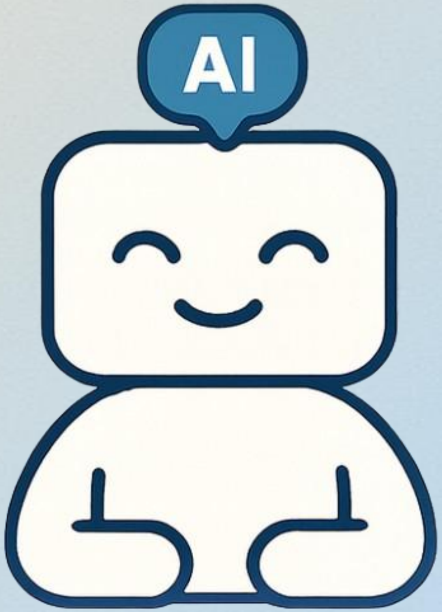
Job Application Service

Personalized, Explainable, Data-Driven Job Search



Source code:
github.com/OlgaSkv/job-application-service

Chrome extension:
github.com/OlgaSkv/save-jd-to-sheets

By Olga Skvortsova

The Beginning

In today's world, it's easy to apply for a job.
AI helps write résumés, tailor keywords, and apply with one click.
It looks effortless — until it's not.

The Reality

Behind that simplicity hides a maze:
Dozens of portals, hundreds of applications, countless automated rejections.

The Mismatch

LinkedIn says: 'You’re a perfect fit!'
ATS says: 'Overqualified. No relevant experience.'

The Struggle

Job search becomes a job of its own — exhausting and demotivating.
I've been there. I followed all the 'right' steps… and still got nowhere.

The Realization

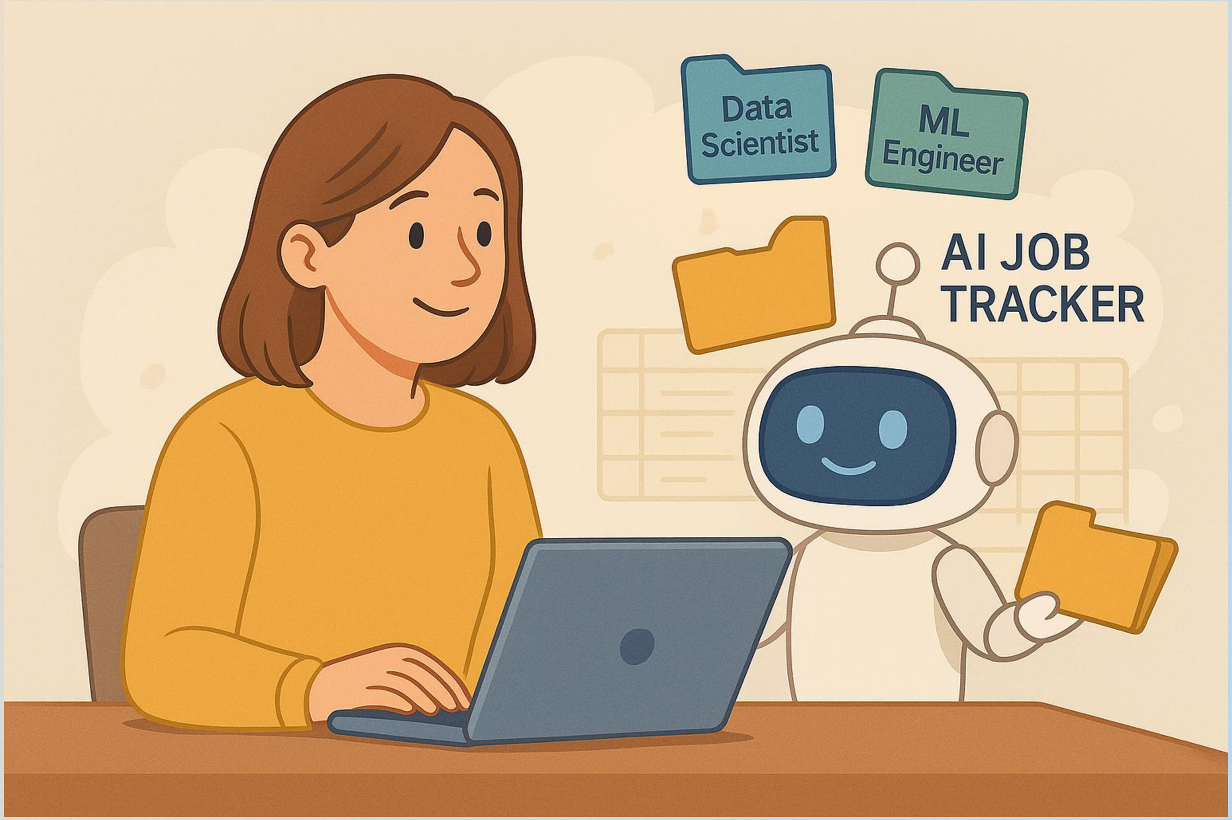
One day, I looked at my spreadsheet — 90 applications,
all expired links and generic titles. They all blurred together.

The Bigger Question

How do you know which jobs to apply for?
Portals flood you with suggestions — not answers.

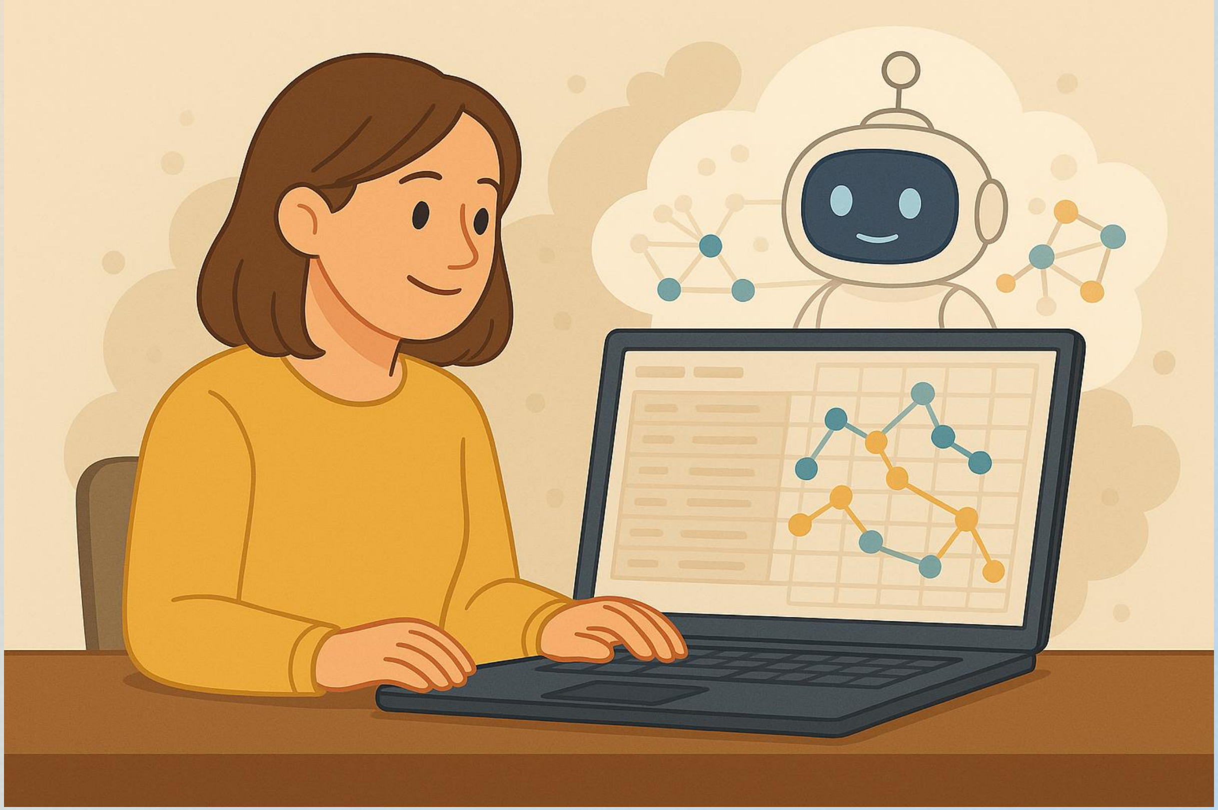
The Spark

That's when the idea was born:
An AI Job Tracker Assistant that could analyze and store every job I applied for.



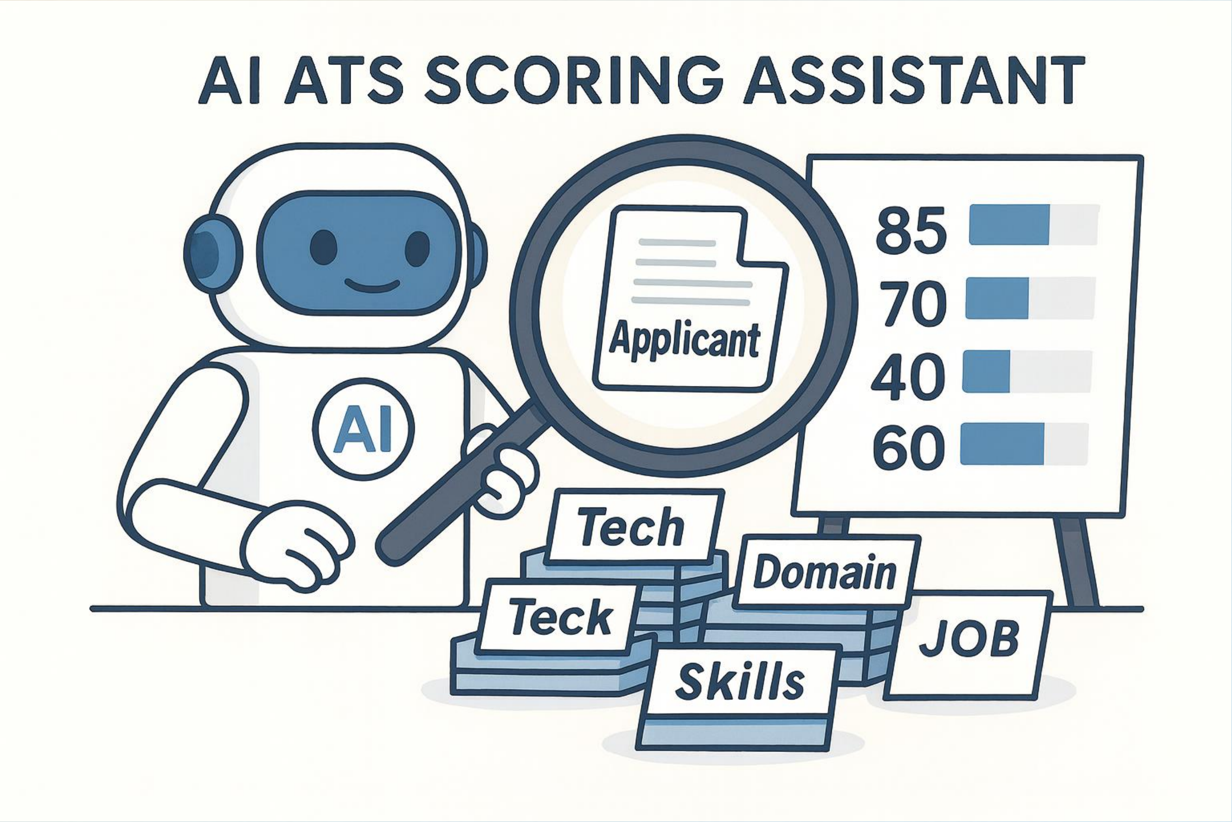
The Data Insight

Imagine 90 applications turned into a dataset — ready for clustering, pattern discovery, and insights.



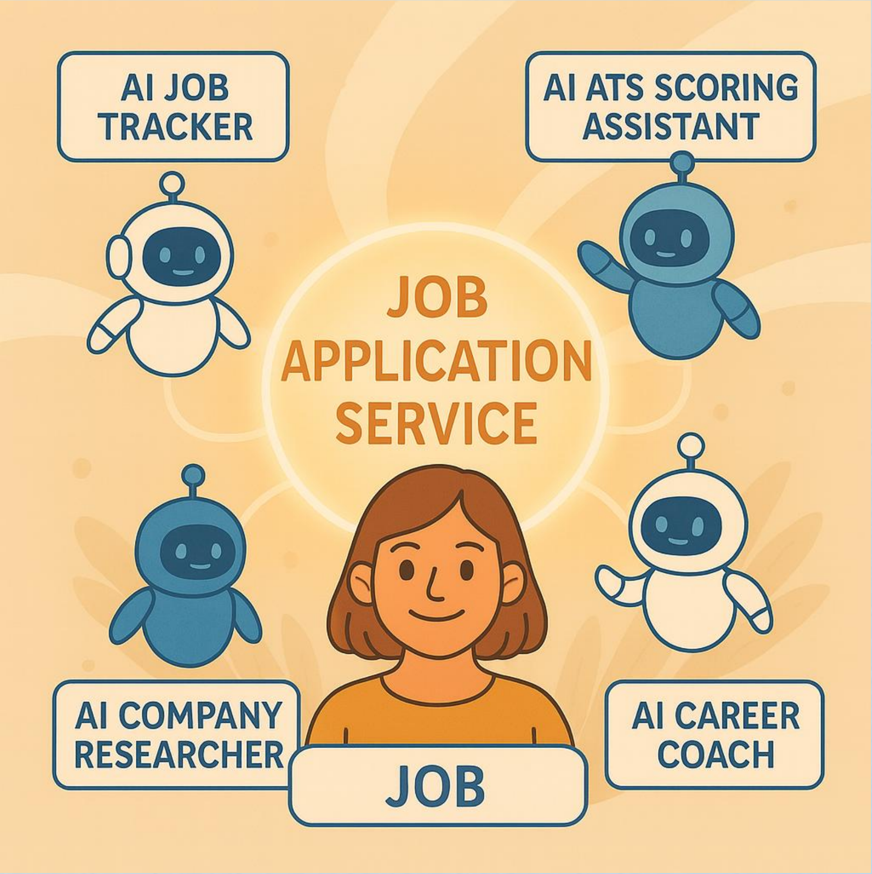
The New Assistant

Meet the AI Scoring Assistant — it evaluates your résumé against the job description, just like an ATS, but transparently.



The Idea Evolves

That's how the Job Application Service was born — not just to analyze jobs, but to understand you.

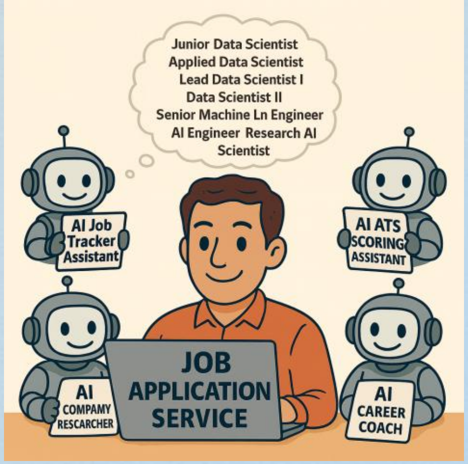


The Solution
From a Spreadsheet… to a System

The Job Application Service transforms job search into a guided, data-driven process. It helps you:
– Score and track each job automatically
– Surface the most relevant keywords from *your résumé* for every platform
– Act as your personal career coach — reviewing each role in your unique context

It helps you decide:
Should you apply? Add a cover letter?
What to highlight — in your own voice.

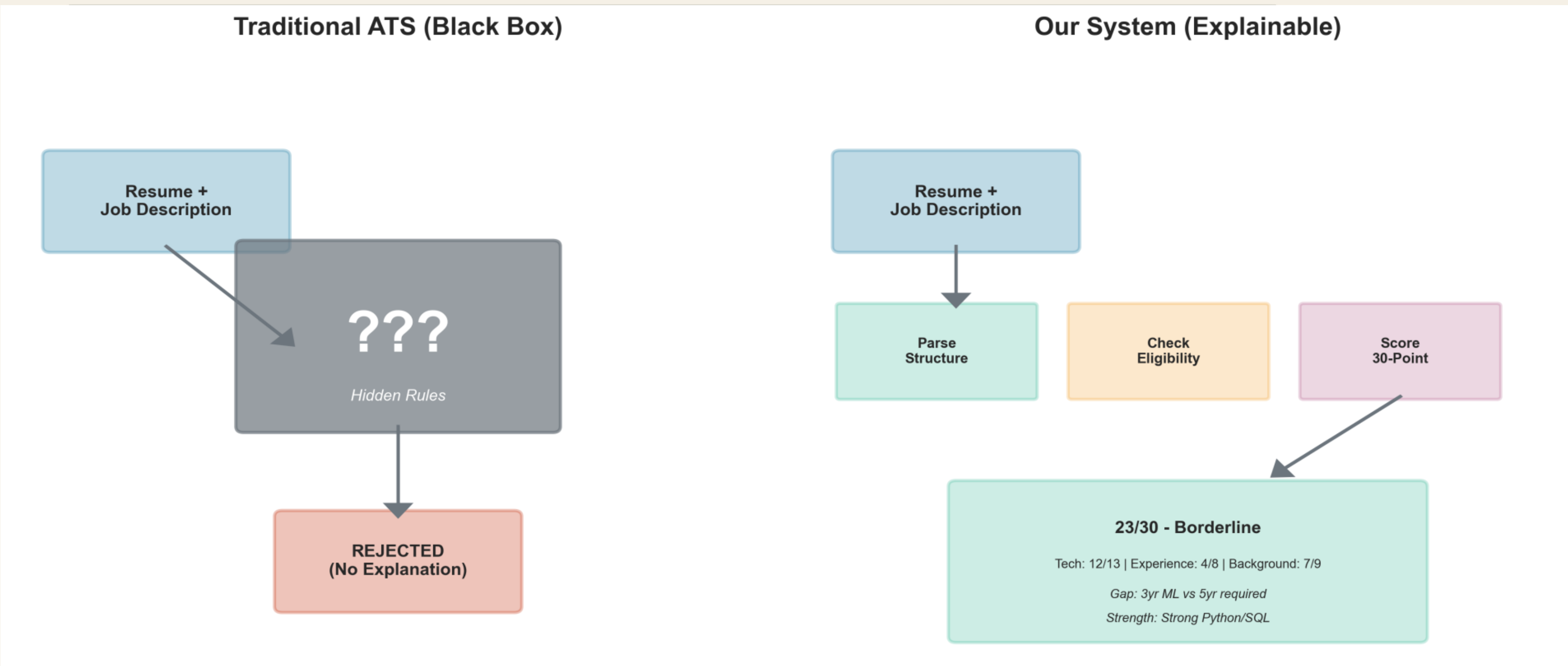
What started as frustration — tens of applications lost in a spreadsheet — became a data problem.



Because smart decisions start with transparent data and clear insight.

Problem & Vision

- Job search remains fragmented and inefficient, driven by unstructured data.
- AI résumé tools optimize surface-level matching instead of meaningful fit.
- ATS systems act as black boxes, rejecting candidates without transparency.
- This project reframes job search as a data science and automation challenge.



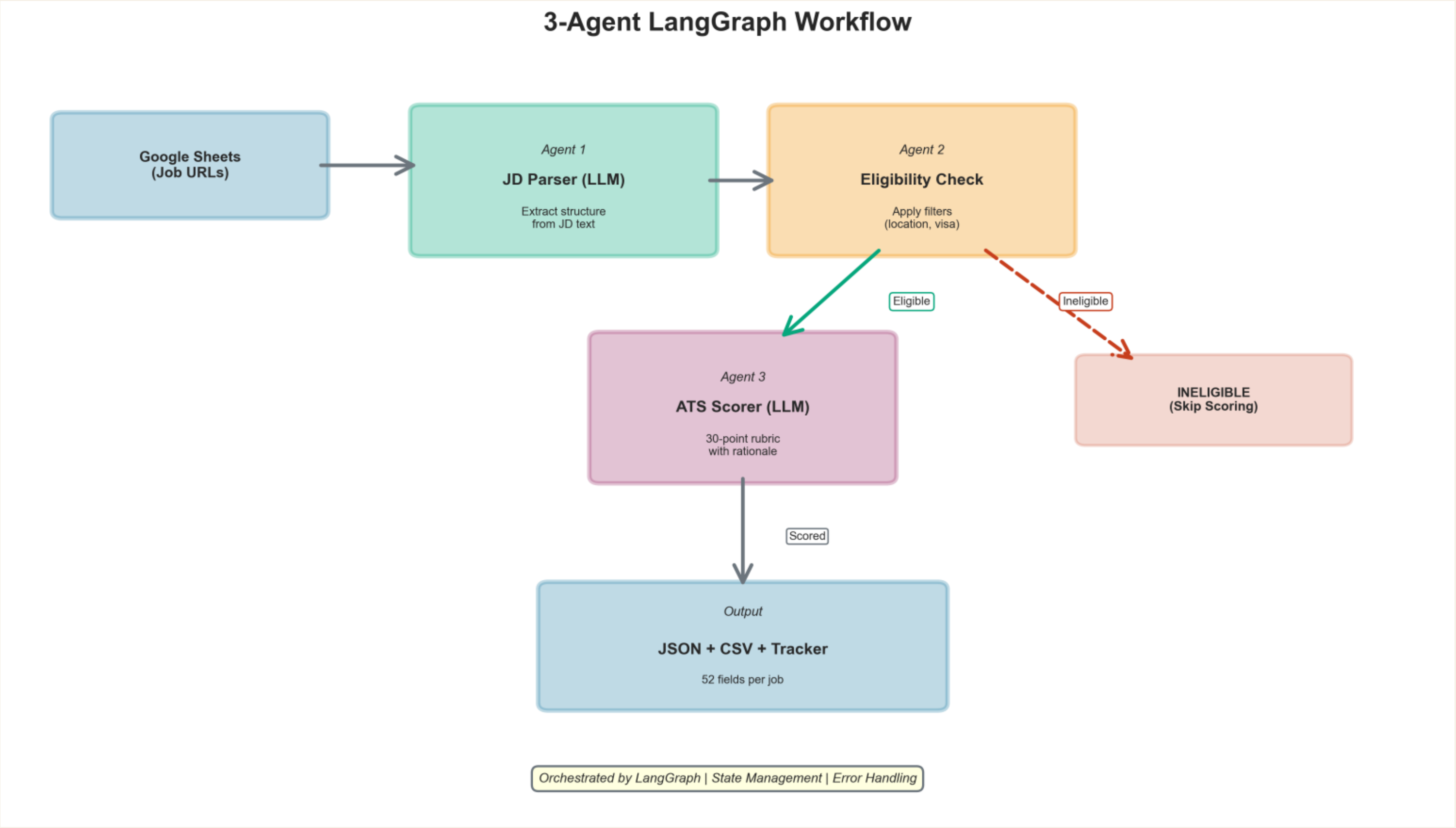
System Architecture

Three AI agents orchestrated via LangGraph: JD Parser, Eligibility Checker, ATS Scorer.

Each agent outputs validated JSON data for downstream steps.

Workflow integrates with Google Sheets for intake and tracking.

Architecture enables modularity, scalability, and transparency.

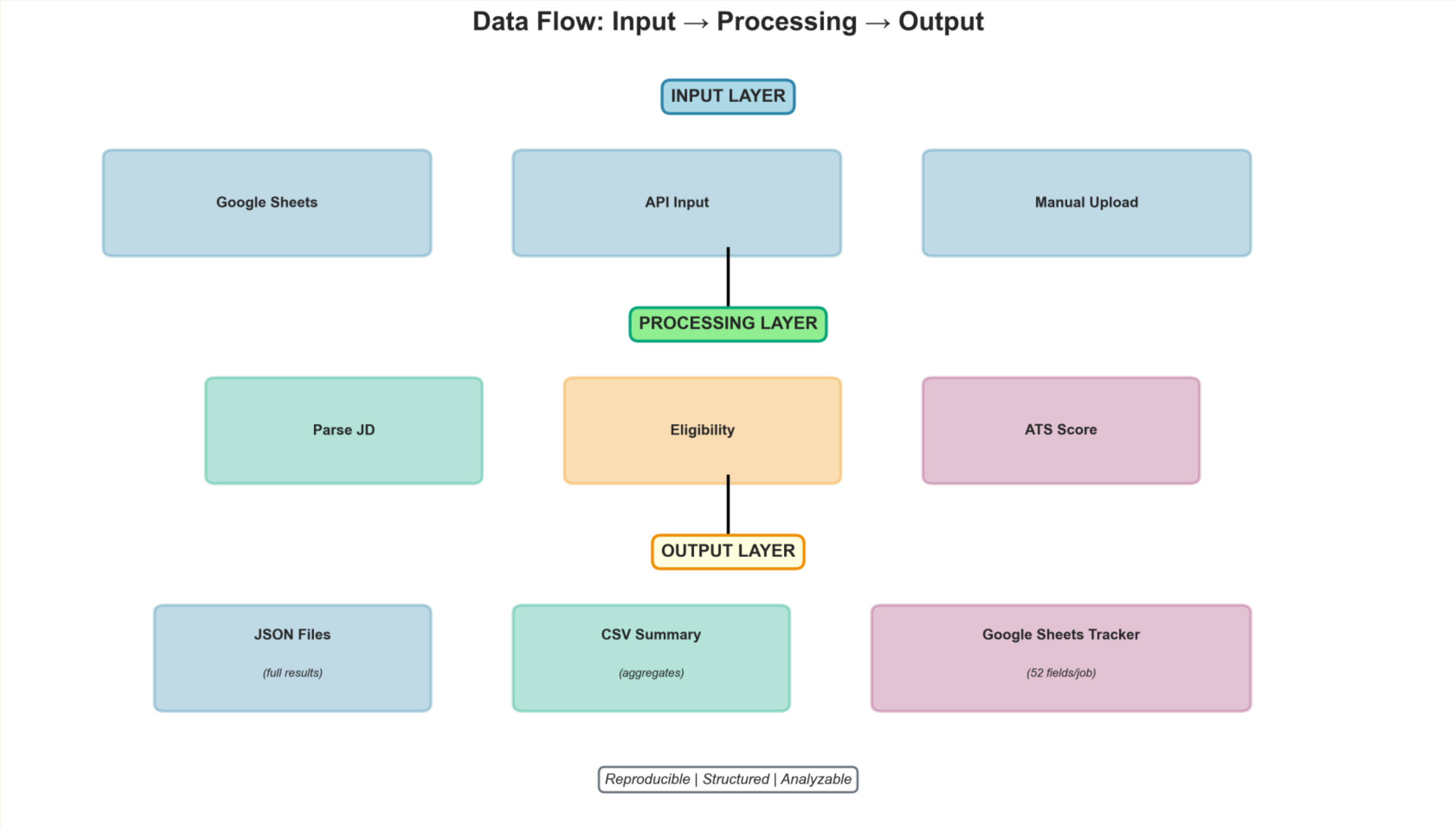


Data Flow

Input: Job descriptions imported from Google Sheets or APIs.

Processing: Structured parsing → Eligibility → Scoring.

Output: JSON results, CSV summaries, and tracker updates.



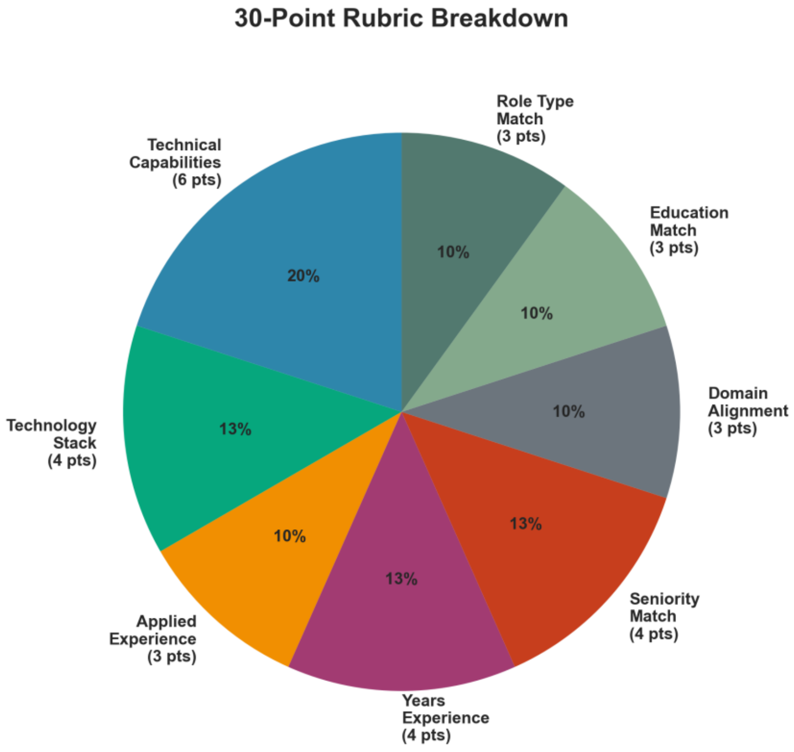
Methodology and Rubric

30-point rubric defining transparent evaluation:

• Technical (13 pts): Skills, tools, applied methods.

• Experience (8 pts): Seniority and duration alignment.

• Background (9 pts): Education and domain match.

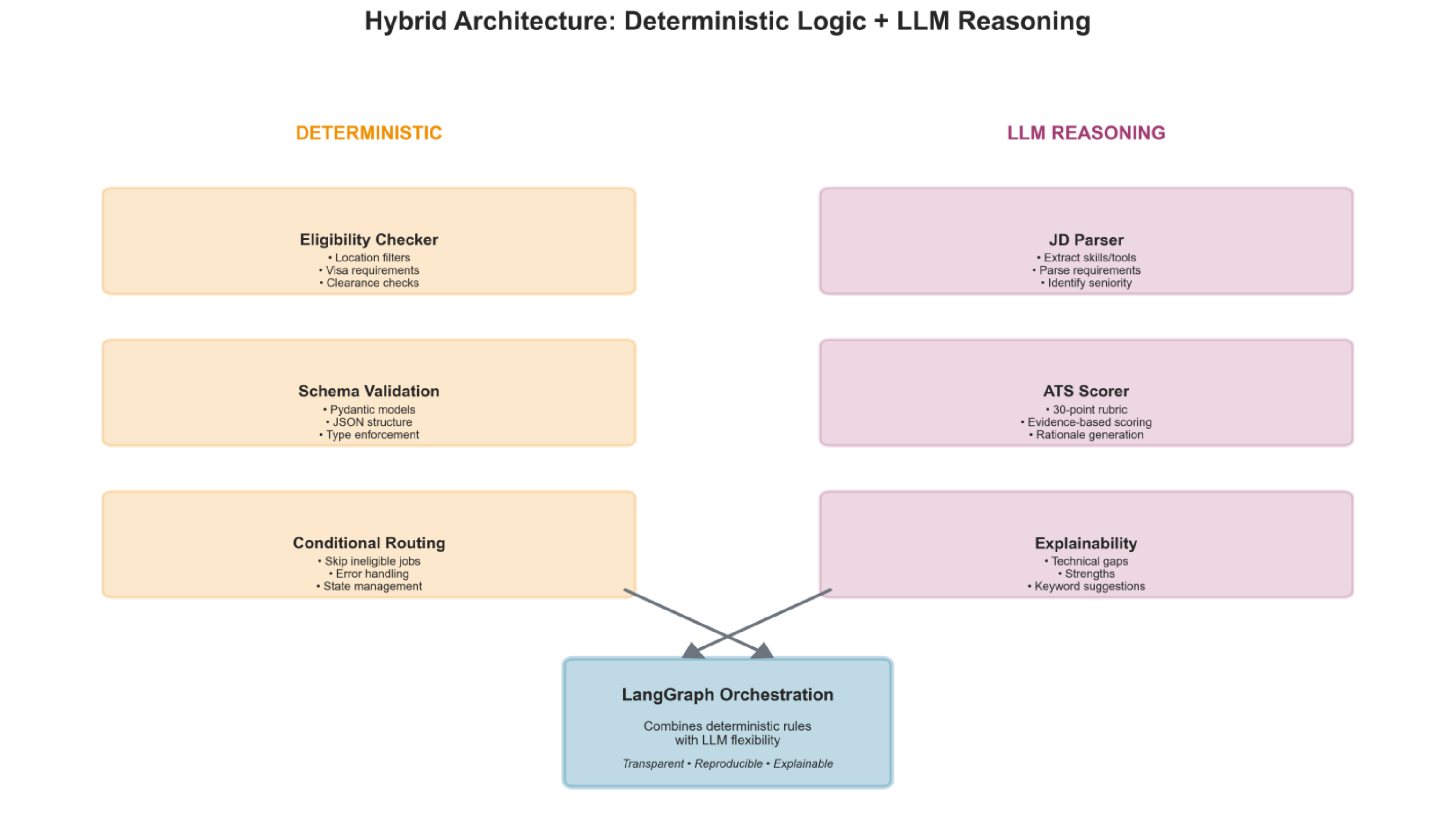


Sample Job Score

Category	Score	Max	Rationale
Technical Capabilities	6	6	All ML methods present
Technology Stack	3	4	Missing Databricks
Applied Experience	3	3	Direct ML deployment
Years Experience	3	4	3yr vs 5yr required (0.6x)
Seniority Match	2	4	Level 3 vs Level 1 (gap=2)
Domain Alignment	2	3	Adjacent industry
Education Match	3	3	Master's exceeds BS req
Role Type Match	3	3	Same role (ML Dev)
TOTAL	**25**	**30**	**ATS PASS (83%)**

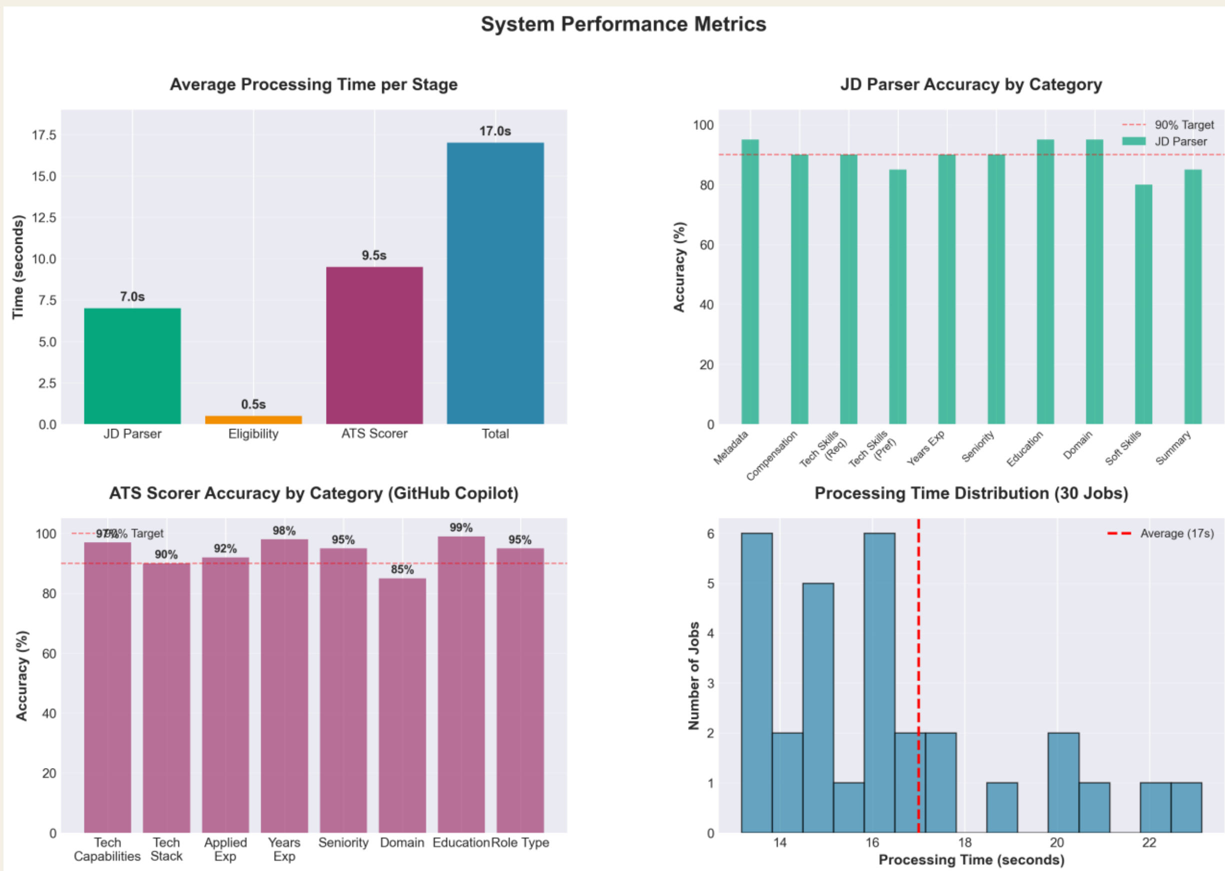
Technical Innovation

- ✓ Hybrid architecture combining deterministic logic and LLM reasoning.
- ✓ Schema-based explainability ensures consistent, interpretable outputs.
- ✓ Conditional routing improves efficiency and reduces cost.
- ✓ LangGraph orchestration enables modular scaling and fault tolerance.



Evaluation and Performance

- Average processing time: 17 seconds per eligible job. Fastest job: 13 s | Slowest: 27 s.
- Average cost per job: $0.02–$0.03 (1-page JD, 1–2 page résumé, 2 LLM agents).
- Accuracy: JD Parser – 95.95 % | ATS Scorer – 89.5 %
- 32 jobs processed (30 eligible) in under 9 minutes end-to-end. LLM cross-evaluation performed using GPT-5 (ChatGPT o1-preview) and Claude 4.5 Sonnet.

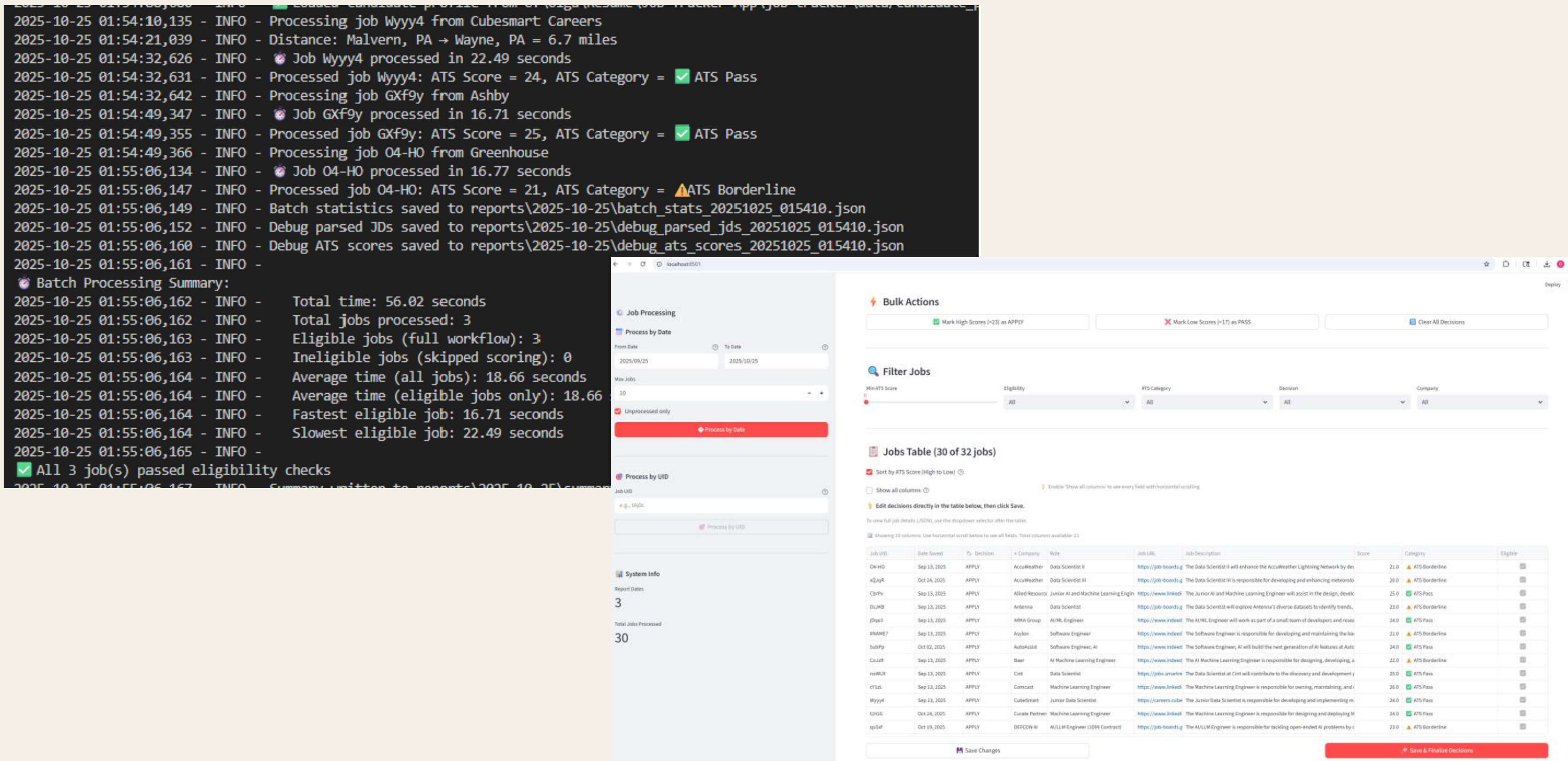


User Interface

Dual interface: automated command-line pipeline and Streamlit dashboard.

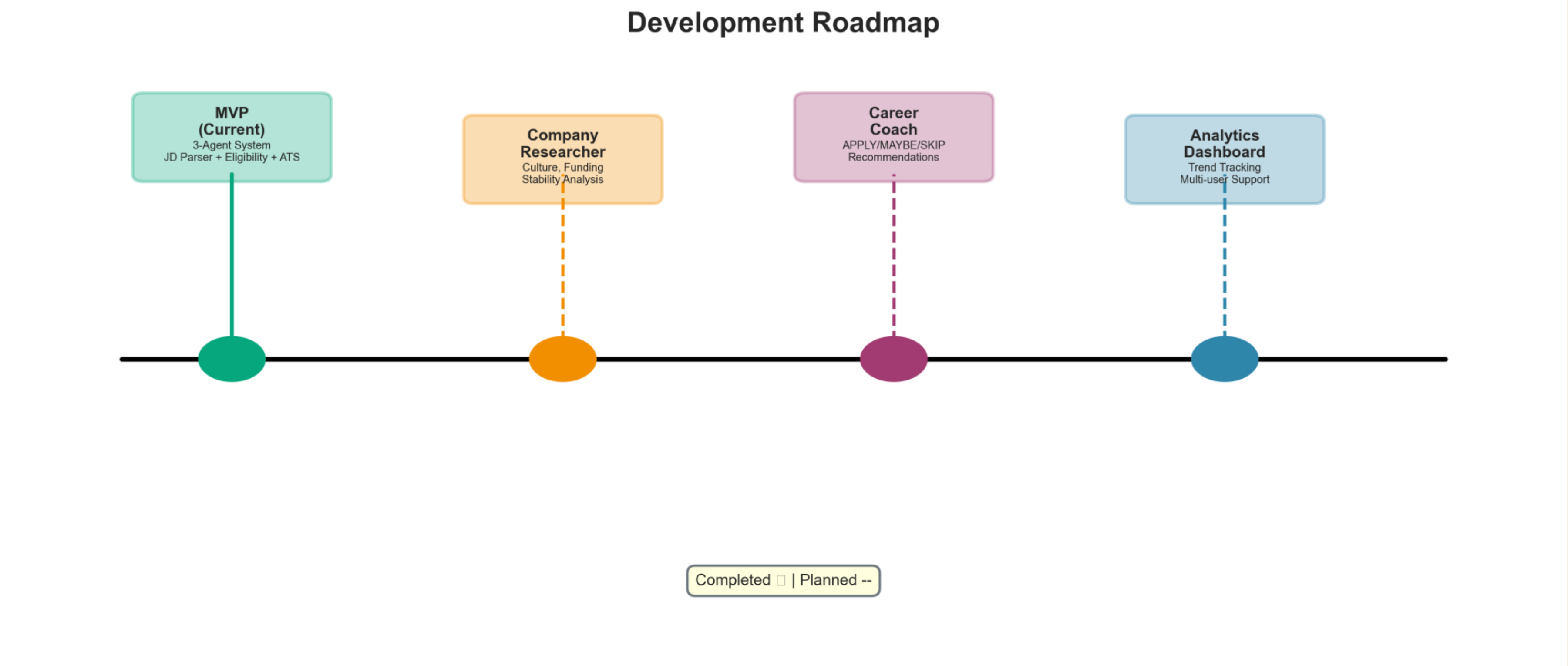
Interactive visualization enables job review, filtering, and transparency.

Focus on human-readable analytics and decision support.



Next Steps

- Expand to 4-agent system: Company Researcher and Career Coach.
- Add dashboards for job analytics and trend visualization.
- Integrate the DeepEval framework for structured and reproducible LLM testing.



Closing Thoughts

- Bridging human intuition and machine reasoning in job search.
- Turning applications into structured, measurable insights.
- Transforming rejections into data for improvement.
- Building toward transparent, adaptive AI-driven career guidance.